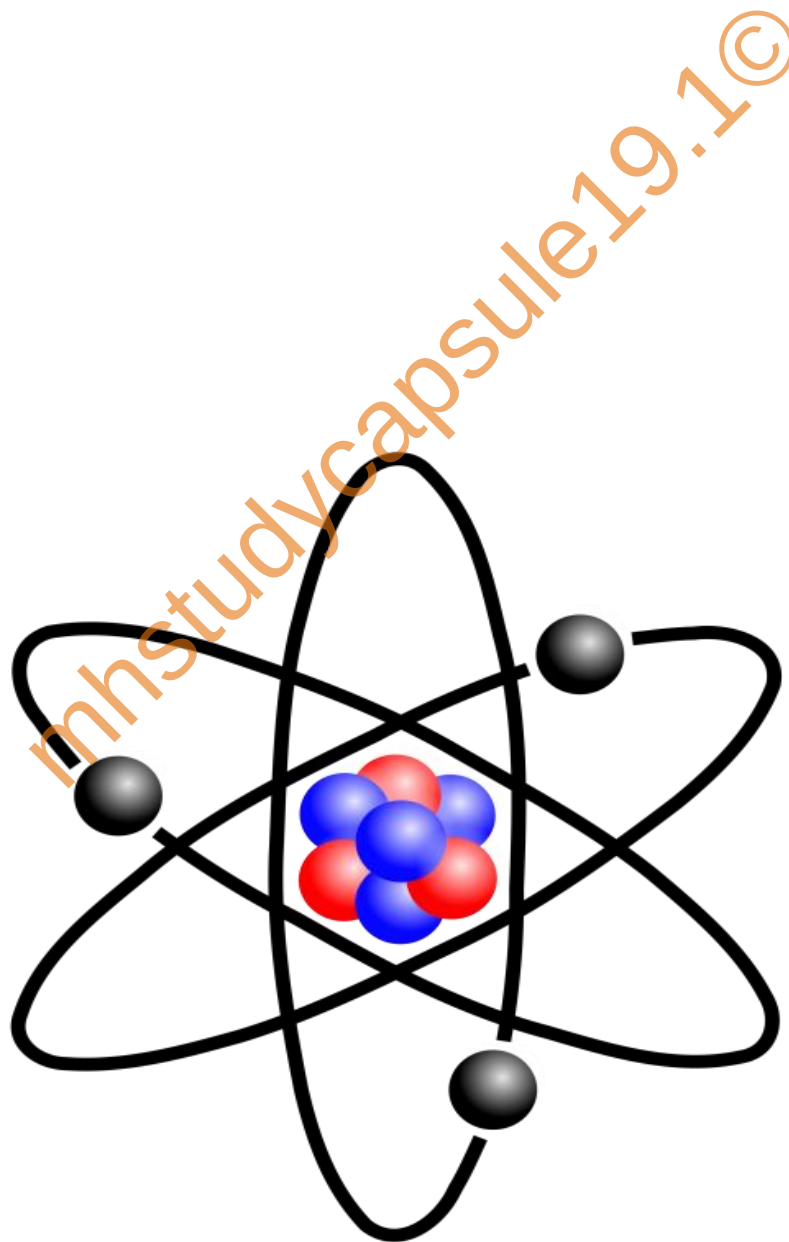


Chapter 3

Chemical Reactions and Equations



1. Choose the correct option from the bracket and EXPLAIN the statement giving reason.

(Oxidation, displacement, electrolysis, reduction, zinc, copper, double displacement, decomposition)

- a. To prevent rusting, a layer of zinc metal is applied on iron sheets.
- b. The conversion of ferrous sulphate to ferric sulphate is oxidation reaction.
- c. When electric current is passed through acidulated water electrolysis of water takes place.
- d. Addition of an aqueous solution of ZnSO_4 to an aqueous solution of BaCl_2 form a white precipitate is an example of double displacement reaction.

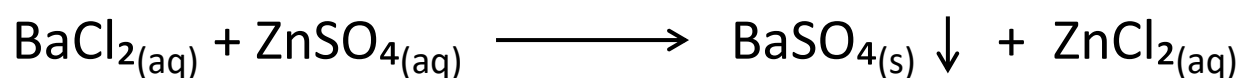
EXPLANATION:

- a. A thin layer of zinc is applied to prevent corrosion of iron. This process is called as Galvanization. Corrosion of zinc occurs first because zinc is more electropositive than iron.
- b. The process of loss of electrons is called as oxidation. When ferrous ions are converted into ferric ions there is loss of one electron.
Reaction for this is $\Rightarrow \text{Fe}^{2+} \longrightarrow \text{Fe}^{3+} + \text{e}^-$ (oxidation).
- c. When electric current is passed through acidulated water, electrolysis of water takes place. Here, hydrogen gas is evolved at cathode and oxygen gas is evolved at anode.

Reaction at cathode $\Rightarrow 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{H}_2$

Reaction at the anode $\Rightarrow 4\text{OH}^- \longrightarrow 2\text{H}_2\text{O} + \text{O}_2$

- d. Addition of aqueous solution of ZnSO_4 to an aqueous solution of BaCl_2 produces ZnCl_2 and BaSO_4 .

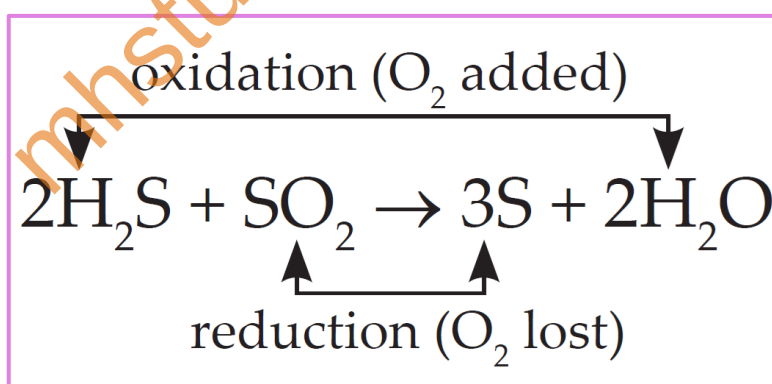


In this reaction ions in the reactants are exchanged to form a precipitate of BaSO_4 . Thus, it is an example of double displacement reaction.

2. Write answers to the following.

- a. What is the reaction called when oxidation and reduction take place simultaneously? Explain with one example.

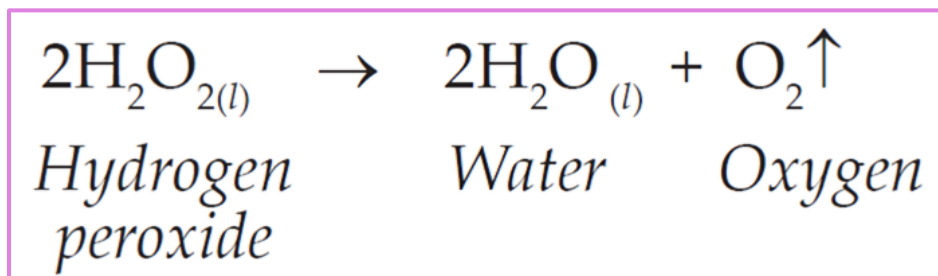
- i. When oxidation and reduction take place simultaneously in a reaction it is termed as Redox reaction.
- ii. During oxidation, a reactant combines with oxygen or loses hydrogen and during reduction it gains hydrogen or loses oxygen.
- iii. Example of Redox reaction:



- iv. In the above reaction, SO_2 is reduced and H_2S is oxidised.

- b. How can the rate of the chemical reaction namely decomposition of hydrogen peroxide be increased?

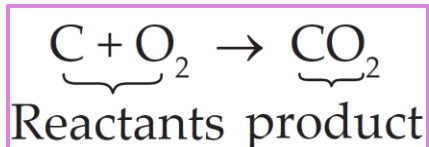
- i. The chemical reaction in which two or more products are formed from a single reactant is called decomposition reaction.
- ii. Hydrogen peroxide decomposes into water and oxygen.



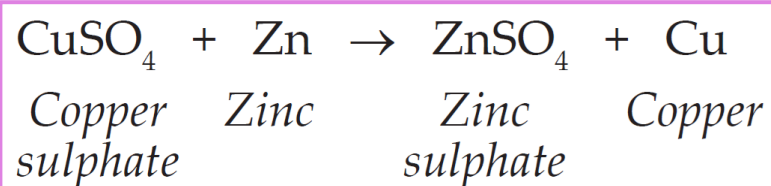
- ii. The decomposition of hydrogen peroxide into water and oxygen takes place slowly at room temperature.
- iii. However, the rate of this chemical reaction can be increased by adding manganese dioxide (MnO_2) powder in it.
- iv. Here, manganese dioxide acts as a catalyst.

c. Explain the term reactant and product giving examples.

- i. A chemical reaction is a process in which some substances undergo bond breaking and are transformed into new substances by formation of new bonds.
- ii. The substances taking part in chemical reaction are called reactants, whereas the substances formed as a result of the chemical reaction by formation of new bonds are called products.
- iii. For example :
 - a. Formation of carbon dioxide gas by combustion of coal in air is a chemical reaction. In this reaction Coal (carbon) and Oxygen (from air) are the reactants while carbon dioxide is the product.



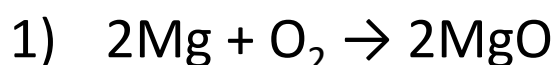
- b. In the given equation, copper sulphate and zinc are the reactants and zinc sulphate and copper are the products.



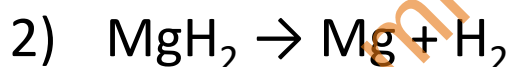
- d. **Explain the types of reaction with reference to oxygen and hydrogen. Illustrate with examples.**

Types of reaction with reference to oxygen and hydrogen are oxidation reaction and reduction reaction.

- i. **Oxidation reaction:** The chemical reaction in which a reactant combines with oxygen or loses hydrogen to form the product is called Oxidation reaction.

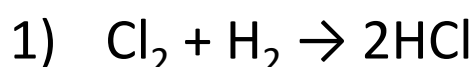


Here O_2 combines with magnesium (Mg) to form magnesium oxide (MgO).

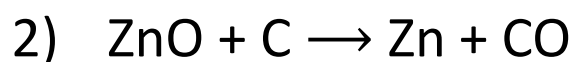


Here, MgH_2 loses hydrogen.

- ii. **Reduction reaction:** The chemical reaction in which a reactant combines with hydrogen or loses oxygen to form product is called reduction reaction.



Here chlorine combines with hydrogen to form hydrochloric acid (HCl).

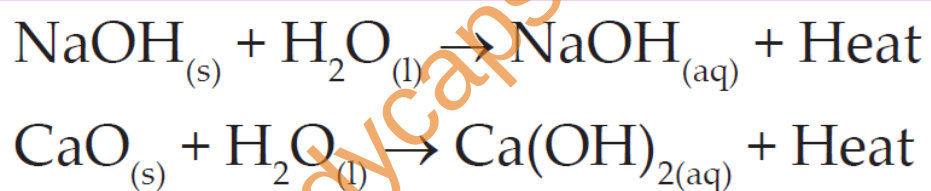


Here, zinc oxide loses oxygen and is reduced to give zinc metal.

e. Explain the similarity and difference in two events, namely adding NaOH to water and adding CaO to water.

1. Similarity:

- i. Both NaOH and CaO dissolve in water. NaOH dissolves in water forming aqueous NaOH while CaO reacts with water to form calcium hydroxide $[\text{Ca}(\text{OH})_2]$
- ii. Both, aqueous NaOH and $\text{Ca}(\text{OH})_2$ are bitter in taste and turn red litmus to blue. i.e., both are basic in nature.
- iii. Both events are exothermic. It means a lot of heat is evolved.



2. Difference:

- i. Sodium Hydroxide (NaOH) is highly soluble in water, whereas calcium oxide (CaO) is less soluble in water.
- ii. NaOH is a monoacidic base, whereas $\text{Ca}(\text{OH})_2$ is a diacidic base.
- iii. As NaOH is highly soluble in water, makes strong base whereas $\text{Ca}(\text{OH})_2$ is a weak base.

3. Explain the following terms with examples.

a. Endothermic reaction

- i. A reaction in which heat is either absorbed from the surrounding or has to be supplied continuously from outside is called as endothermic reaction.
- ii. For example, decomposition of calcium carbonate to produce calcium oxide and carbon dioxide requires continuous heating from outside.
- iii. $\text{CaCO}_3(s) + \text{heat} \longrightarrow \text{CaO}(s) + \text{CO}_2(g)$

b. Combination reaction

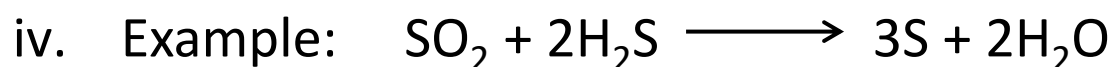
- i. When two or more reactants combine in a reaction to form a single product, it is called a combination reaction.
- ii. Example: When a Magnesium strip is burnt in air, it combines with oxygen, to form white ash of magnesium oxide.
- iii. $2\text{Mg}_{(s)} + \text{O}_{2(g)} \longrightarrow 2\text{MgO}_{(s)} \text{ (white ash)}$

Magnesium

Magnesium oxide

c. Balanced equation

- i. The number of atoms of elements in the reactants is same as the number of atoms of those elements in the products. Such an equation is called a 'balanced equation'.
- ii. In any reaction, the total mass of each of the respective elements in the reactants is same as the total mass of each of the respective elements in the products.
- iii. This is in accordance with the law of conservation of mass.



On reactant side

No. of atoms of Sulphur (S) = 3

No. of atoms of Oxygen (O) = 2

No. of atoms of hydrogen (H) = 4

On product side

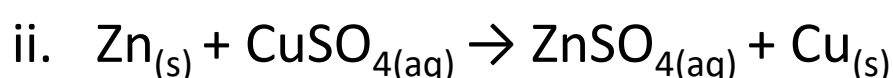
No. of atoms of Sulphur (S) = 3

No. of atoms of Oxygen (O) = 2

No. of atoms of hydrogen (H) = 4

d. Displacement reaction

- i. The reaction in which the place of the ion of a less reactive element in a compound is taken by another more reactive element by formation of its own ions, is called displacement reaction.



Zinc

*Copper
sulphate*

*Zinc
sulphate*

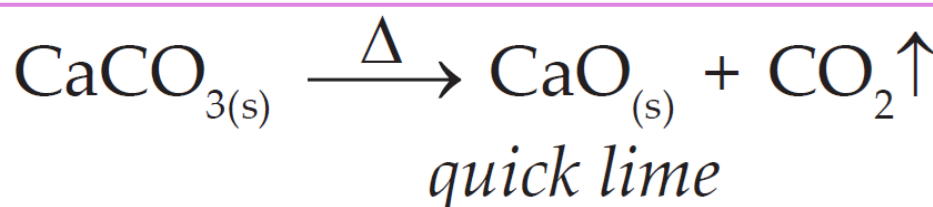
Copper

- iii. In the above equation when zinc dust is added to blue coloured copper sulphate solution, a colourless solution of zinc sulphate is formed and copper is deposited on zinc.

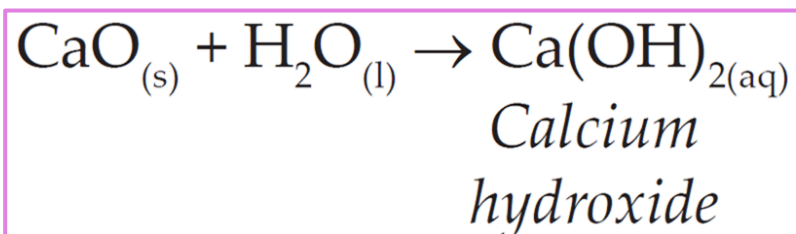
4. Give scientific reasons.

- a. When the gas formed on heating limestone is passed through freshly prepared lime water, the lime water turns milky.

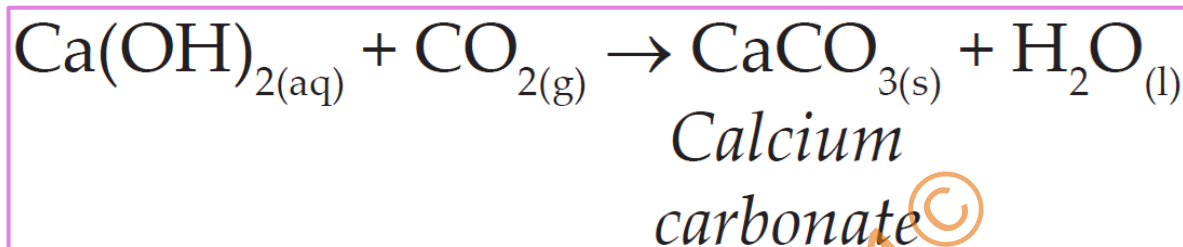
- i. On heating, limestone undergoes thermal decomposition to form calcium oxide (quick lime) and carbon dioxide.



- ii. Calcium oxide when dissolved in water forms Calcium hydroxide (lime water).



- iii. When carbon dioxide is passed through lime water it turns milky due to formation of white precipitate of Calcium carbonate.



b. It takes time for pieces of Shahabad tiles to disappear in HCl, but its powder disappears rapidly.

- i. When HCl is added to Shahabad stone effervescence of CO_2 is formed.
- ii. The CO_2 effervescence is formed slowly with the pieces of Shahabad tiles while at the faster speed with the powder.
- iii. This is because the rate of a reaction depends upon the size of the particles of the reactants taking part in the reaction.
- iv. Smaller the size of the reactant particles more is the surface area available for the reaction and higher is the rate of reaction.
- v. Hence, it takes time for pieces of Shahabad tiles to disappear in HCl, but its powder disappears rapidly.

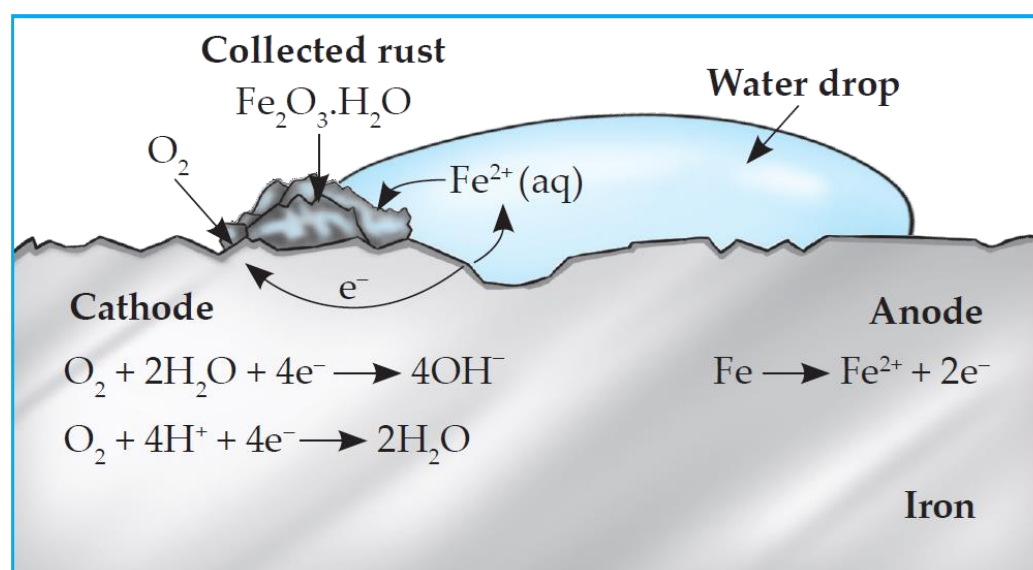
c. While preparing dilute sulphuric acid from concentrated sulphuric acid in the laboratory, the concentrated sulphuric acid is added slowly to water with constant stirring.

- i. Concentrated sulphuric acid (H_2SO_4) is a hazardous, highly corrosive chemical. In the process of dilution of concentrated sulphuric acid with water, very large amount of heat is liberated.
- ii. As a result, water gets evaporated instantaneously along with splashing of acid which may cause an accident.
- iii. To avoid this, required amount of water is taken in a glass container and small quantity of concentrated sulphuric acid at a time is added with stirring.
- iv. Therefore, only a small amount of heat is liberated at a time.

d. It is recommended to use airtight container for storing oil for long time.

- i. When we use old, left over cooking oil for making food stuff, it is found to have foul odour called rancidity. If food is cooked in such oil, its taste also changes.
- ii. When oil or ghee is left aside for a long time or fried food is left aside for a long time it undergoes air oxidation and becomes rancid.
- iii. The oxidation process is greatly enhanced when edible oils are stored in containers that are not airtight and in areas where oil is exposed to air, heat and light.
- iv. Hence, it is recommended to use airtight containers for storing oil for long time, to prevent it from getting oxidised.

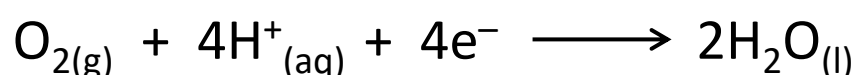
5. Observe the following picture and write down the chemical reaction with explanation.



- In the above diagram, process of rusting takes place.
- A certain type of reddish coloured solid layer collects on the metallic surface.
- This layer is called rust.
- Its chemical formula is $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$. The rust is formed by an electrochemical reaction.
- Different regions on the surface of iron become anode and cathode.
- Fe is oxidised to Fe^{2+} in the anode region.

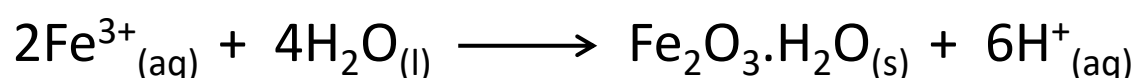


- O_2 reduces to form water in the cathode region.



- When Fe^{2+} ions migrate from the anode region, they react with water and further get oxidised to form Fe^{3+} ions.

- ix. A reddish coloured hydrated oxide is formed from Fe^{3+} ions, it is called rust. It collects on the surface.



- x. Due to various components of atmosphere, oxidation of metals takes place, consequently resulting in their damage. This is called Corrosion.
- xi. Iron rusts and a reddish coloured layer is collected on it. This is corrosion of iron.

6. Identify from the following reactions the reactants that undergo oxidation and reduction.



In the reaction, Iron combines with Sulphur to form FeS. That means, Fe loses electrons to form Fe^{+2} and S gains electrons to form S^{-2} . Loss of electron from a substance is called oxidation, while gain of electrons is called reduction. So, Iron undergoes oxidation while Sulphur undergoes reduction.



In the reaction, silver oxide decomposes to produce silver and oxygen. That is, oxygen is being removed from silver oxide. Removal of oxygen from a substance is called reduction, so silver oxide undergoes reduction.

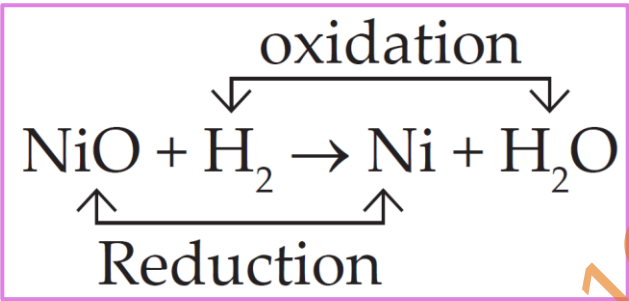


In the reaction, magnesium combines with oxygen to form magnesium oxide. That means, oxygen is being added to magnesium.

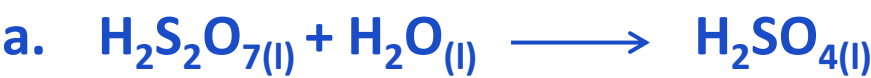
Addition of oxygen to a substance is called oxidation, so magnesium undergoes oxidation.



In the above equation, reactant NiO (Nickel oxide) loses oxygen, thus undergoes reduction whereas H₂ (Hydrogen gas) gains oxygen forming water and undergoes oxidation.



7. Balance the following equation stepwise.



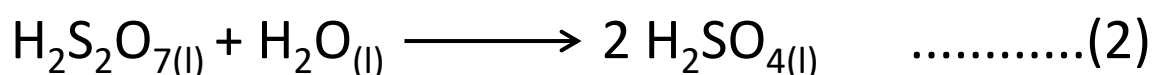
Step 1: Write the equation and then count the number of atoms of each element on the R.H.S. and L.H.S. of the arrow.



Elements	No. of atoms in reactants (L.H.S)	No. of atoms in products (R.H.S)
H	4	2
S	2	1
O	8	4

Step 2: Check whether no. of atoms on both sides is equal. It is seen from the above table number of atoms of all the elements on the two sides are not the same.

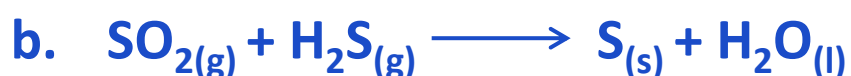
Step 3: To balance the number of atoms on both sides, we write coefficient 2 for H_2SO_4 . Now the equation becomes.



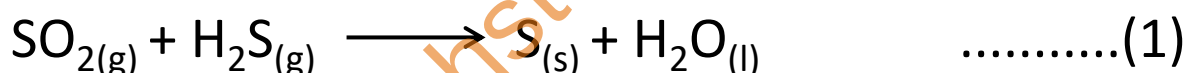
Now see that the number of atoms of each element are equal on both sides.

Elements	No. of atoms in reactants (L.H.S)	No. of atoms in products (R.H.S)
H	4	4
S	2	2
O	8	8

Hence, the equation (2) is a balanced equation.



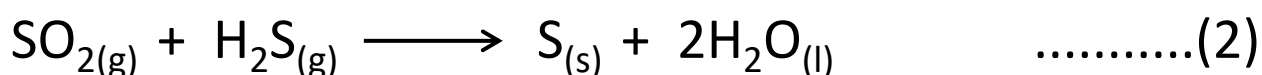
Step 1: Write the equation and then count the number of atoms of each element on the R.H.S. and L.H.S. of the arrow.



Elements	No. of atoms in reactants (L.H.S)	No. of atoms in products (R.H.S)
H	2	2
S	2	1
O	2	1

Step 2: Check whether no. of atoms on both sides is equal. It is seen from the above table number of atoms of all the elements on the two sides are not the same.

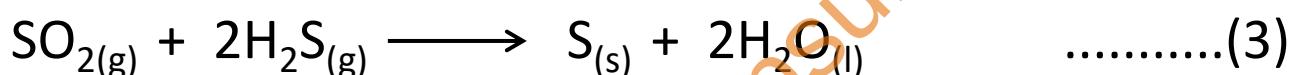
Step 3: We will start with oxygen. On product side we need 2 atoms of oxygen, hence coefficient 2 for H₂O. Now the equation will be



Again, count the number of atoms of elements on both the sides.

Elements	No. of atoms in reactants (L.H.S)	No. of atoms in products (R.H.S)
H	2	4
S	2	1
O	2	2

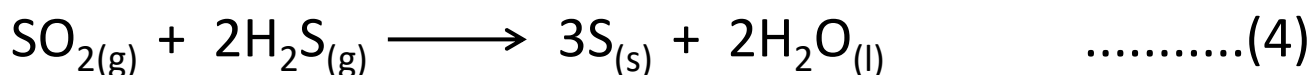
Still the equation is unbalanced. To balance sulphur and hydrogen atoms on reactant side, write coefficient 2 for H₂S



Again, count the number of atoms of elements on both the sides.

Elements	No. of atoms in reactants (L.H.S)	No. of atoms in products (R.H.S)
H	4	4
S	3	1
O	2	2

Now if we count number of atoms of hydrogen and oxygen, they are equal. But number of atoms of sulphur is not balanced. Now there are 3 atoms of sulphur on reactant side, whereas on product sides it is only one atom. Hence, we write coefficient 3 for sulphur on R.H.S. Now the equation becomes.



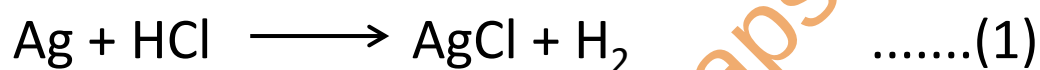
Again, count the number of atoms of elements on both the sides.

Elements	No. of atoms in reactants (L.H.S)	No. of atoms in products (R.H.S)
H	4	4
S	3	3
O	2	2

It means equation (4) is a balanced equation.

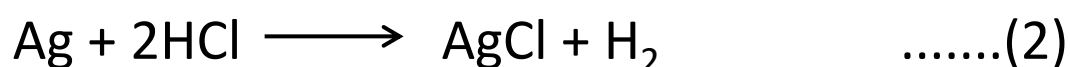


Step 1: Write the chemical equation and count the numbers of atoms in reactants and products.



Elements	No. of atoms in reactants (L.H.S)	No. of atoms in products (R.H.S)
Ag	1	1
Cl	1	1
H	1	2

Step 2: From the above table, we see that the number of atoms of hydrogen are balanced. So, to balance hydrogen, we write coefficient 2 for HCl on L.H.S.



Again, count the number of atoms of elements on both sides.

Elements	No. of atoms in reactants (L.H.S)	No. of atoms in products (R.H.S)
Ag	1	1
Cl	2	1
H	2	2

Step 3: Now to balance chlorine, we write coefficient 2 for AgCl on R.H.S.



Again, count the number of atoms of elements on both sides.

Elements	No. of atoms in reactants (L.H.S)	No. of atoms in products (R.H.S)
Ag	1	2
Cl	2	2
H	2	2

Step 4: Atoms of elements, hydrogen and chlorine are now balanced. Number of atoms of Ag on R.H.S is 2, and on L.H.S is 1. To balance number of atoms of Ag on L.H.S, we write coefficient 2 for Ag.



Again, count the number of atoms of elements on both sides.

Elements	No. of atoms in reactants (L.H.S)	No. of atoms in products (R.H.S)
Ag	2	2
Cl	2	2
H	2	2

It means equation (4) is a balanced equation.



Step 1: Write the chemical equation and count the numbers of atoms in reactants and products.



Elements	No. of atoms in reactants (L.H.S)	No. of atoms in products (R.H.S)
Na	1	2
O	5	5
H	3	2
S	1	1

The number of oxygen and sulphur atoms are equal on both sides. Hence, we will balance Na atoms on both sides

Step 2: To balance Na atom we write coefficient 2 for NaOH.



Count the number of Na atoms, H atoms, O atoms and S atoms on both sides.

Elements	No. of atoms in reactants (L.H.S)	No. of atoms in products (R.H.S)
Na	2	2
O	6	5
H	4	2
S		

Step 3: To balance H, we write coefficient 2 for H₂O on the product side.

Hence the equation becomes.



Again, count the number of atoms of elements on both sides.

Elements	No. of atoms in reactants (L.H.S)	No. of atoms in products (R.H.S)
Na	2	2
O	6	6
H	4	4
S	1	1

Now the number of atoms of elements are equal on both sides. Hence, equation (3) is a balanced equation.

8. Identify the endothermic and exothermic reaction.



Heat is liberated in the product side. So, it is an exothermic reaction.



Heat is supplied continuously from outside for the decomposition reaction of potassium chlorate to take place. So, It is an endothermic reaction.



Heat is liberated in the product side. So, it is an exothermic reaction.



Heat is supplied continuously from outside for the decomposition reaction of calcium carbonate to take place. So, It is an endothermic reaction.

9. Match the column in the following table.

Reactants	Products	Type of chemical reaction
$\text{BaCl}_2 (\text{aq}) + \text{ZnSO}_4 (\text{aq})$	$\text{BaSO}_4 \downarrow + \text{ZnCl}_2 (\text{aq})$	Double displacement
$2\text{AgCl}_{(s)}$	$2\text{Ag}_{(s)} + \text{Cl}_{2(g)}$	Decomposition
$\text{CuSO}_4 (\text{aq}) + \text{Fe}_{(s)}$	$\text{FeSO}_4 (\text{aq}) + \text{Cu}_{(s)}$	Displacement
$\text{H}_2\text{O}_{(l)} + \text{CO}_{2(g)}$	$\text{H}_2\text{CO}_{3(\text{aq})}$	Combination

INTEXT QUESTIONS

1. What are the types of molecules of elements and compounds?

- i. Monoatomic, diatomic, triatomic, and polyatomic molecules of elements are all available.
- ii. Diatomic, triatomic, tetratomic and polyatomic categories can be used to describe compound molecules.

2. What is meant by valency of elements?

The number of electrons accepted, donated or shared by an atom so as to attain noble gas configuration is called valency.

3. What is the requirement for writing molecular formulae of different compounds? How are the molecular formulae of the compounds written?

- i. The things required to write the molecular formula of a compound are the symbols of elements, numbers, and sometimes other things like brackets.
- ii. Molecular formulae of compounds are written using the chemical symbols of the constituent elements. Subscripts are used to denote the number of atoms of each element present. For example, water is represented as H_2O , indicating two hydrogen (H) atoms and one oxygen (O) atom in the molecule.

4. Identify physical and chemical changes from the phenomena given in the following table.

Phenomenon	Physical change	Chemical change
1. Transformation of ice into water.	✓	
2. Cooking of food.		✓
3. Ripening of fruit.		
4. Milk turned in to curd.		
5. Evaporation of water.		
6. Digestion of food in the stomach.		
7. Size reduction of naphtha balls exposed to air.		
8. Staining of Shahbad or Kadappa tile by lemon juice.		
9. Breaking of a glass object on falling from a height.		

5. What are the other uses of silver nitrate in everyday life?

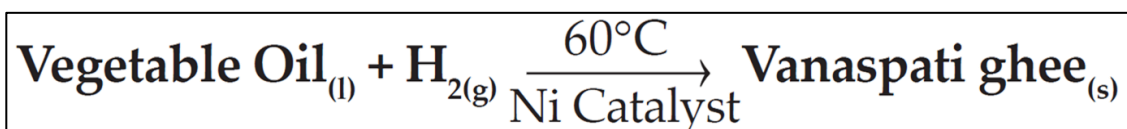
- For the removal of small quantities of unwanted tissues such as warts.
- By dentists, in order to heal mouth ulcers.
- In electroplating.
- In the manufacturing of mirrors.
- In the preparation of dyes and also used in photography

5. Carry out the activities and record observations:

- I. Take a spoonful of limestone powder in an evaporating dish. Heat it strongly on a high blue flame.
- II. Add zinc (Zn) dust into the copper sulphate (CuSO_4) solution.
- III. Add potassium chromate (K_2CrO_4) solution to barium sulphate (BaSO_4) solution.
- IV. Add sodium carbonate (Na_2CO_3) solution to the calcium chloride (CaCl_2) solution.
- V. Take phthalic anhydride in the evaporating dish.

Activity	Colour change (if present)	Gas released (yes/no)	Temperature change (if present)	Nature of change (chemical/physical)
1	No	Yes	No	Chemical
2	Yes	No	Yes. Temp increases	Chemical
3	Yes. Yellow precipitate is formed	No	No	Chemical
4	Yes. White precipitate is formed	No	Yes. Temp increases	Chemical
5	No	Sublimation	No	Physical

6. Identify the reactants and products of equation:



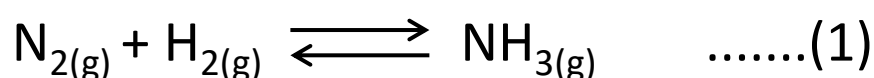
Reactants: Vegetable oil and hydrogen gas

Product: Vanaspati ghee

7. Write down the steps in balancing the equation

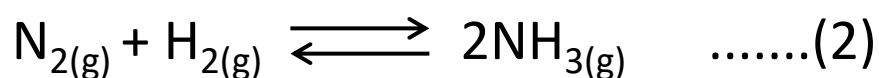


Step 1: Write the chemical equation and count the numbers of atoms in reactants and products.



Elements	No. of atoms in reactants (L.H.S)	No. of atoms in products (R.H.S)
N	2	1
H	2	3

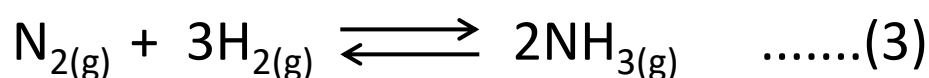
Step 2: Number of atoms of elements is not same on both sides. So, we need to balance it. First, we will balance Nitrogen by writing coefficient 2 for the product NH_3 .



Again, count the number of atoms of elements on both sides.

Elements	No. of atoms in reactants (L.H.S)	No. of atoms in products (R.H.S)
N	2	2
H	2	6

Step 3: Number of atoms of nitrogen is now balanced on both sides, but number of atoms of hydrogen is not balanced. Therefore, we write coefficient 3 for hydrogen on reactant side. The equation becomes:



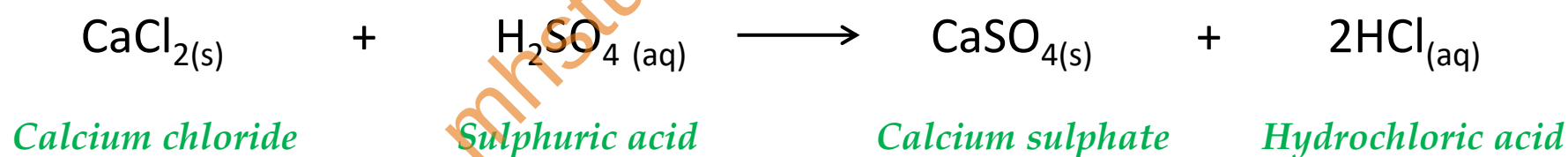
Now, count the number of atoms on both side.

Elements	No. of atoms in reactants (L.H.S)	No. of atoms in products (R.H.S)
N	2	2
H	6	6

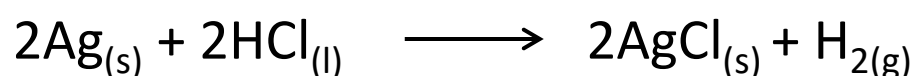
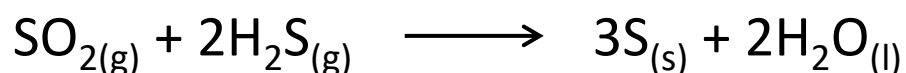
It means the equation (3) is a balanced equation.

8. Write down a balanced chemical equation for the following reaction.

Calcium chloride + Sulphuric acid → Calcium sulphate + Hydrogen chloride



9. Write down the physical states of reactants and products in following reactions.



10. What is the number of reactants in the following reactions.



In all the above equations, there are two reactants.

What is the number of molecules of reactants taking part in the above reactions?



1 molecule *1 molecule*



2 molecule *1 molecule*



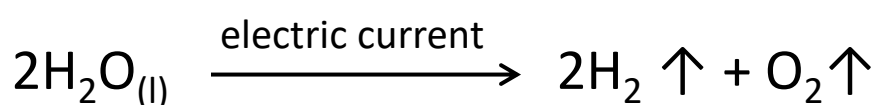
1 molecule *1 molecule*

How many products are formed in each of the above reaction?

In all the above reactions, only one product is formed.

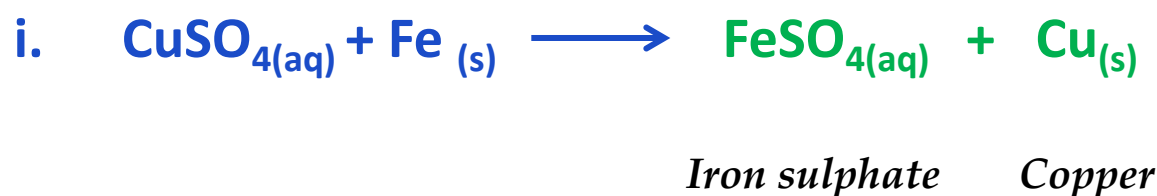
11. Is it possible to produce hydrogen by decomposition of water by means of heat, electricity or light ?

Yes. It is possible to produce hydrogen by decomposition of water. Water decomposes into hydrogen and oxygen on passing electric current through acidulated water. This decomposition takes place by means of electrical energy.



It is called electrolysis.

12. Complete the following reactions and give names of the products.



13. What is the difference in the process of dissolution and a chemical reaction ?

Dissolution refers to the process of dissolving a solute into a solvent to make a solution. In chemical reaction, reactants will react chemically with each other, and new products are formed. The bonds in the reactants are broken and new bonds are formed.

14. Does a new substance form when a solute dissolves in a solvent ?

No. New substance is not formed when a solute dissolves in a solvent.

15. Take into account the time required for following processes. Classify them into two groups and give titles to the groups.

Fast reaction	Slow reaction
Cooking gas starts burning on ignition.	Iron article undergoes rusting.
Effervescence is formed on adding baking soda into a test tube containing dilute acid.	Erosion of rocks takes place to form soil.

A white precipitate is formed on adding dilute sulphuric acid to barium chloride solution.

Alcohol is formed on mixing yeast in glucose solution under proper condition.

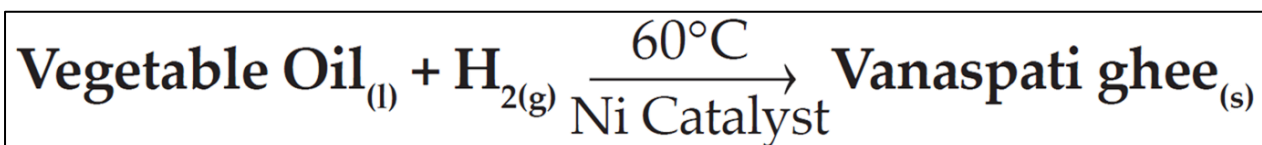
16. Which is the oxidant used for purification of drinking water?

- Chlorine is a strong oxidant commonly used in water treatment for oxidation and disinfection.
- Even Potassium permanganate can be used as an oxidant for purification of water.

17. Why is potassium permanganate used during cleaning water tanks?

- Potassium permanganate oxidises dissolved iron, manganese and hydrogen sulphide turning them into solid particles.
- KMnO_4 is used to remove organic matter from the water tank by oxidation which also removes unwanted taste and odour.
- It also kills harmful bacteria and fungi.

18. Look at the chemical equation



What is the type of this reaction, in which Vanaspathi ghee is formed from vegetable oil?

In the preparation of vanaspathi ghee from vegetable oil hydrogen gas is used. This process is known as hydrogenation. This is reduction reaction.

19. Some examples of redox reaction are as follows. Identify the reductants and oxidants from them.



Here, H_2S is the reductant and SO_2 is the oxidant

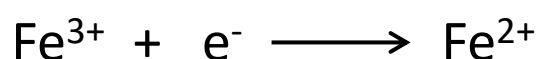


Here, MnO_2 is the oxidant and HCl is the reductant.

20. If oxidation means losing electrons, what is meant by reduction?

Reduction means gaining electrons.

21. Write the reaction of formation of Fe^{2+} by reduction Fe^{3+} by making use of the symbol (e^-).



22. The luster of the surface of the aluminium utensils in the house is lost after a few days. Why does this happen?

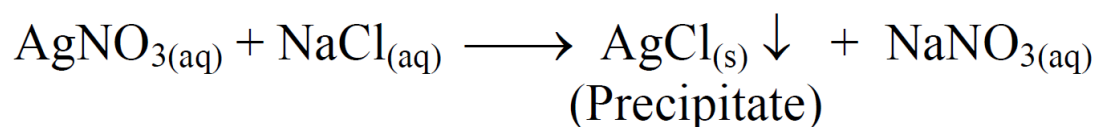
- Aluminium utensils when kept in the house for a few days, undergo oxidation.
- A thin layer aluminium oxide (Al_2O_3) is deposited on the surface.
- Hence, aluminium utensils lose their luster in a few days.

23. How are the blackened silver utensils and patinated (greenish) brass utensils cleaned?

Blackened silver utensils can be cleaned using aluminium foil, hot water, baking soda or salt while patinated (greenish) brass utensils can be cleaned using lemon juice and baking soda.

Previous Year Questions

1. Study the following chemical reaction and answer the questions given below:



- i. Identify and write the type of chemical reaction.

Double displacement

- ii. Write the definition of above type of chemical reaction.

The reaction in which the ions in the reactants are exchanged to form a precipitate is a double displacement reaction.

- iii. Write the names of reactants and products of above reaction.

Reactants – silver nitrate, sodium chloride

Products – silver chloride, sodium nitrate

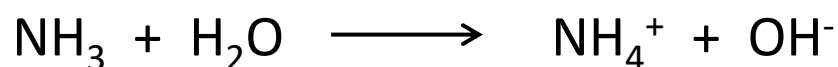
2. Explain the similarity and difference in two events namely adding NaOH to water and adding CaO to water. – **Exercise Question 2 - e**

3. Observe the following reaction carefully and answer the sub-questions:



- a. What are the salt and acid in the above reaction?
- b. Which is the base in the above reaction? It is weak or strong?
- c. Write a reaction showing dissociation of this base in water.

- NH_4Cl is the salt and HCl is the acid.
- Ammonia (NH_3) is the base in the reaction. It is a weak base.
- Dissociation of ammonia in water can be shown as follows:

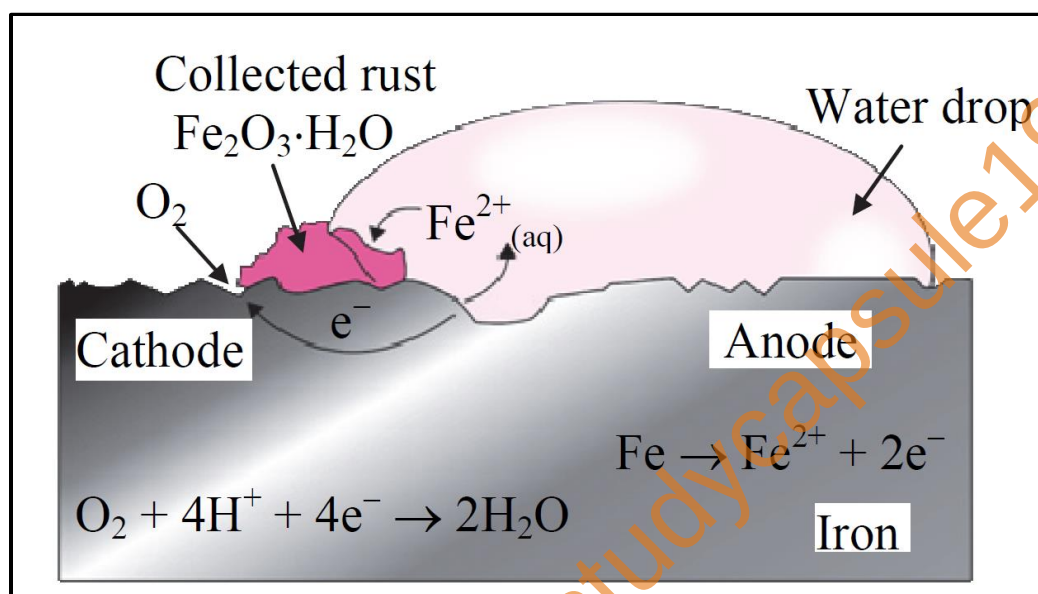


4. Complete the given chemical reaction:



Name the type of the reaction. – Displacement reaction

5. Observe the following picture and answer the following questions:



- What is rust?
- Write the chemical formula of rust.
- Write the reaction of oxidation of iron at anode.
- Write the reaction of oxidation of iron at cathode.
- What is corrosion?

Refer Exercise Question 5

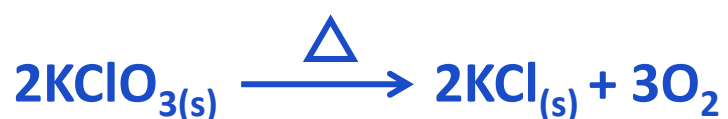
6. Identify 'A' and 'B' from the following table and complete the table.

Write the chemical equation:



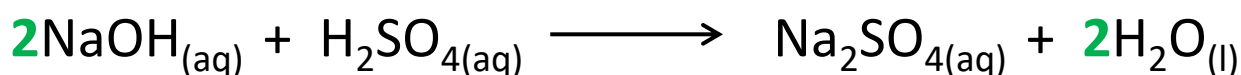
7. Answer the following:

a. Write the 'endothermic' or 'exothermic' nature of the reaction:



Heat is supplied continuously from outside for the decomposition reaction of potassium chlorate to take place. So, It is an endothermic reaction.

b. Balance the given chemical equation:

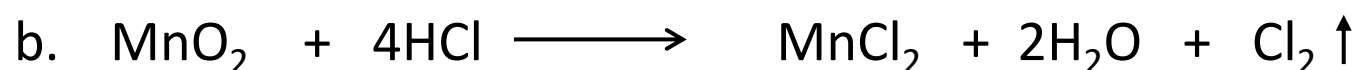
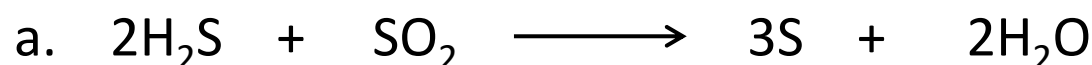
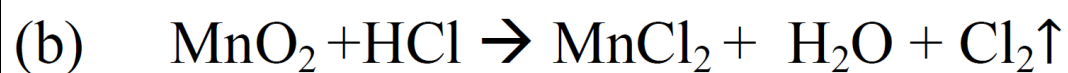
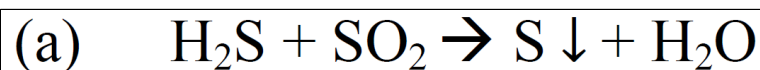


c. From given reaction, identify 'oxidant' and 'reductant':



CuO is the oxidant, H₂ is the reductant.

8. Balance the following chemical equation (Do not write step by step)

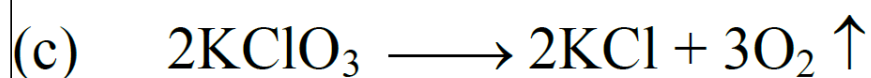
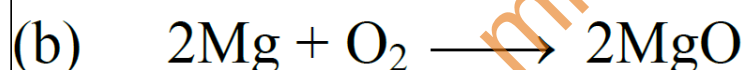
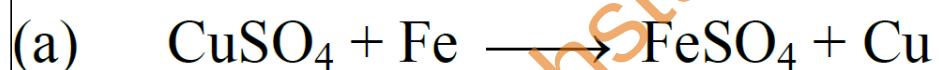


9. (a) What is the chemical formula of rust?

(b) Write the chemical reaction of electric current passing through positive and negative surface of iron. – Refer Exercise Question 5

10. It is recommended to use airtight container for storing oil for long time. – Exercise Question 4 - d

11. Identify the type of chemical reaction given below:



a. Displacement

b. Combination

c. Decomposition

12. Complete the following reaction:

12 C

11 H₂O

13. Identify the physical and chemical changes from the following phenomena:

- (a) Transformation of ice into water - **Physical**
- (b) Ripening of fruit - **Chemical**
- (c) Milk turned into curd - **Chemical**
- (d) Evaporation of water - **Physical**
- (e) Digestion of food in the stomach - **Chemical**
- (f) Iron fillings get attracted towards the magnet - **Physical**

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